Part A

1. Provide a short explanation of how you implemented the questiongeneration methods

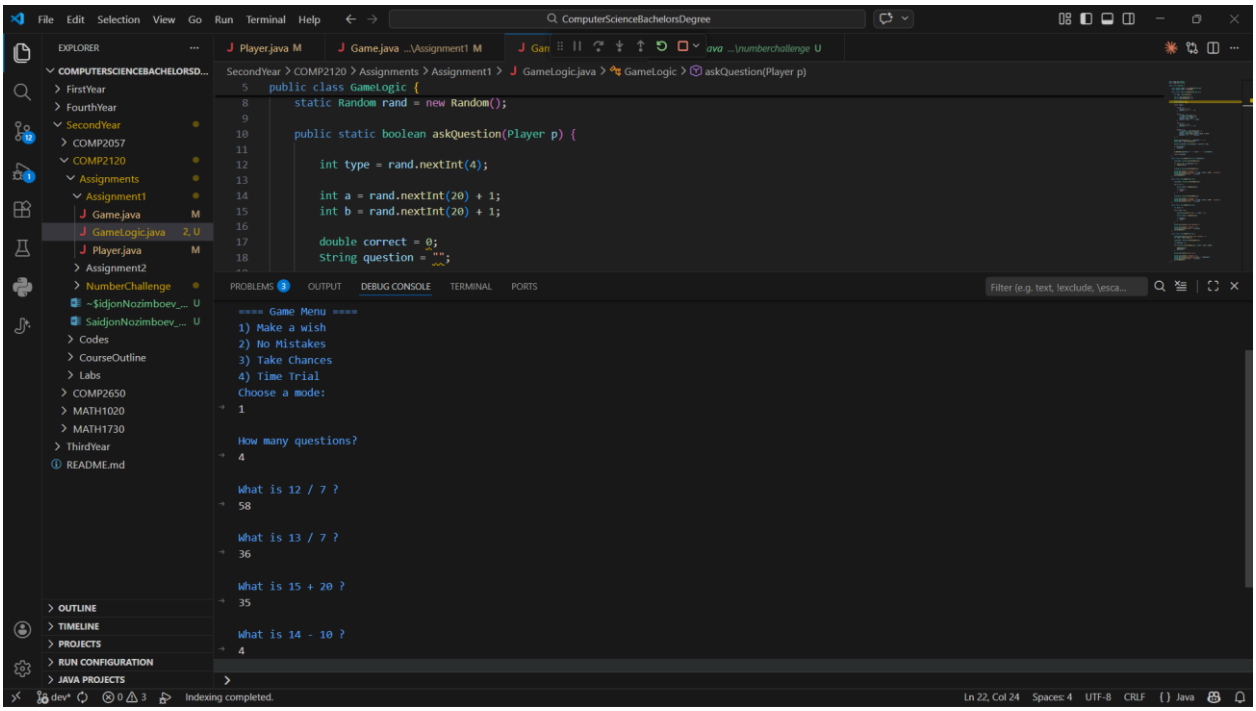
I generated two random numbers and random number for operations(+ - * /) then used switch case to implement operations and calculate

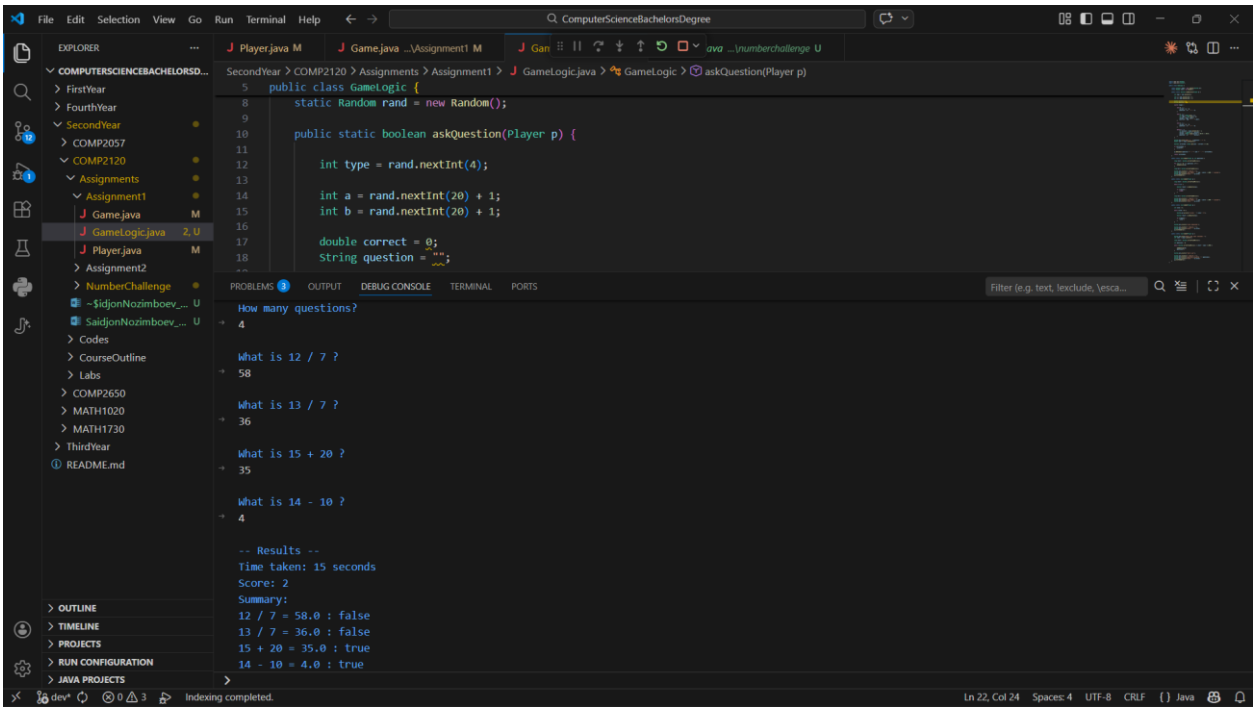
```java
public class GameLogic {

    public static boolean askQuestion(Player p) {

        int type = rand.nextInt(4);

        int a = rand.nextInt(20) + 1;
        int b = rand.nextInt(20) + 1;

        double correct = 0;
        String question = "";

        switch (type) {

            case 0 -> {
                correct = a + b;
                question = a + " + " + b;
            }

            case 1 -> {
                int big = Math.max(a, b);
                int small = Math.min(a, b);
                correct = big - small;
                question = big + " - " + small;
            }

            case 2 -> {
                correct = a * b;
                question = a + " * " + b;
            }

            default -> {
                int divisor = rand.nextInt(10) + 1;
                correct = (double) a / divisor;
                correct = Math.round(correct * 100.0) / 100.0;
                question = a + " / " + divisor;
            }
        }
```

```java
public class GameLogic {
    public static boolean askQuestion(Player p) {
                correct = a + b;
                question = a + " + " + b;
            }

            case 1 -> {
                int big = Math.max(a, b);
                int small = Math.min(a, b);
                correct = big - small;
                question = big + " - " + small;
            }

            case 2 -> {
                correct = a * b;
                question = a + " * " + b;
            }

            default -> {
                int divisor = rand.nextInt(10) + 1;
                correct = (double) a / divisor;
                correct = Math.round(correct * 100.0) / 100.0;
                question = a + " / " + divisor;
            }
        }

        System.out.print("What is " + question + " ? ");
        double user = input.nextDouble();

        boolean correctAns = Math.abs(user - correct) < 0.01;

        if (correctAns) {
            p.score++;
        }

        p.addSummary(question + " = " + user + " : " + correctAns);

        return correctAns;
    }
```

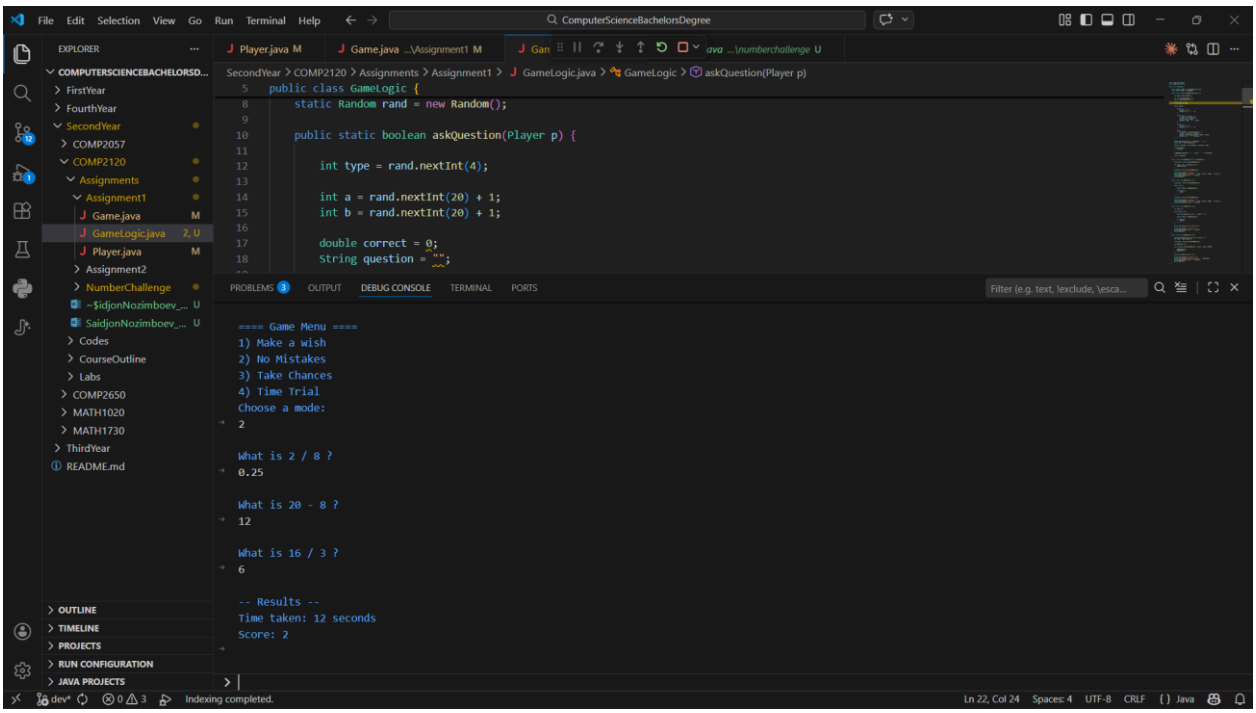
2. Program Execution Screenshots

Mode1

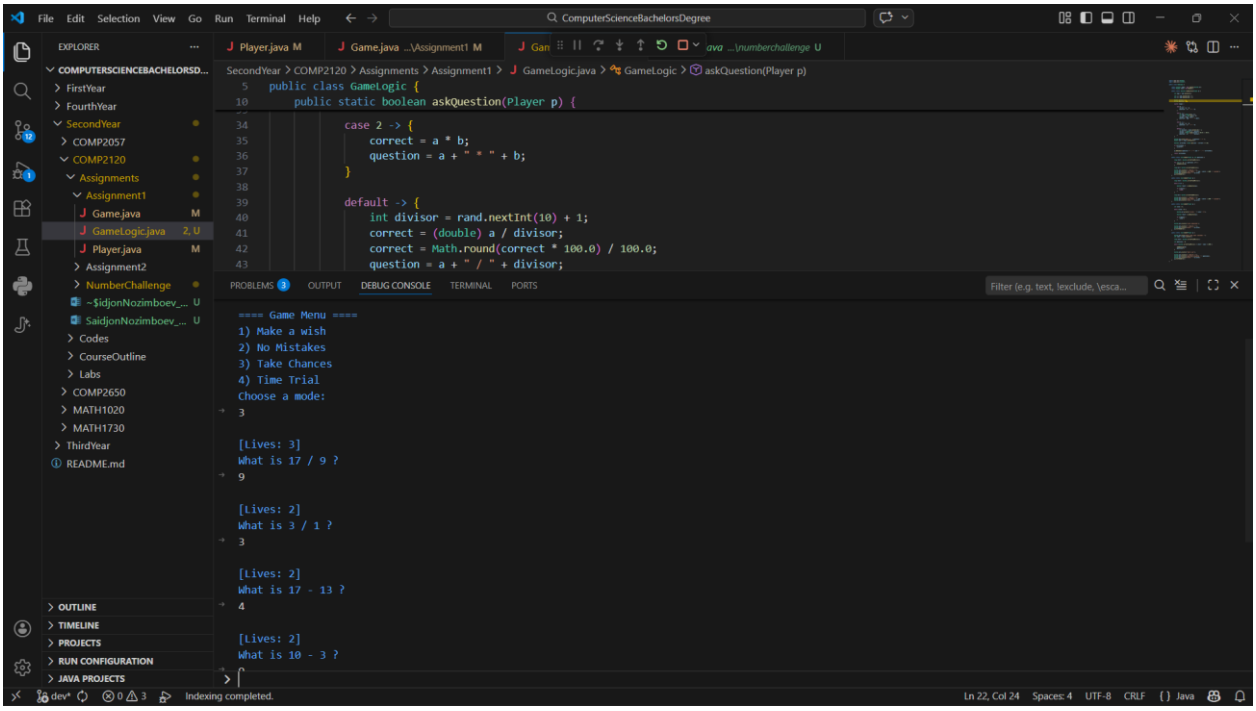


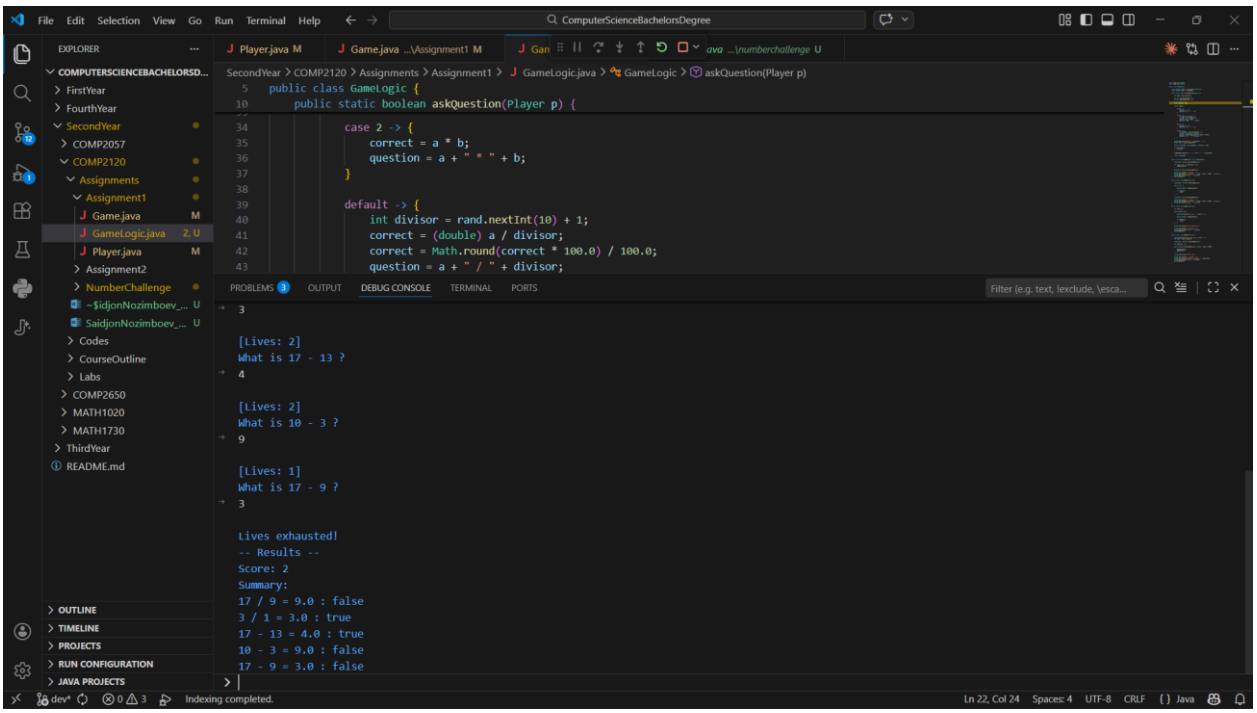


Mode2

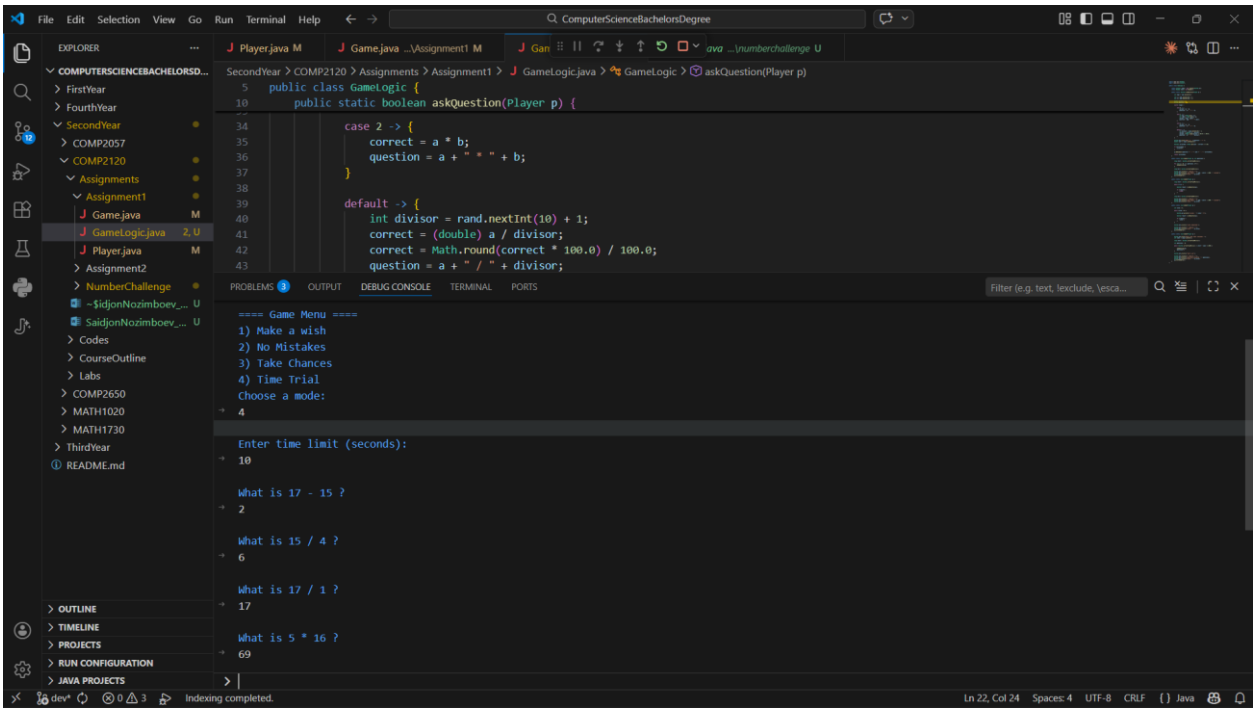

SecondYear > COMP2120 > Assignments > Assignment1 > GameLogic.java > GameLogic > askQuestion(Player p)
public class GameLogic {
static Random rand = new Random();
public static boolean askQuestion(Player p) {
int type = rand.nextInt(4);
int a = rand.nextInt(20) + 1;
int b = rand.nextInt(20) + 1;
double correct = 0;
String question = "";
==== Game Menu ====
1) Make a wish
2) No Mistakes
3) Take Chances
4) Time Trial
Choose a mode:
2
What is 2 / 8 ?
0.25
What is 20 - 8 ?
12
What is 16 / 3 ?
6
-- Results --
Time taken: 12 seconds
Score: 2

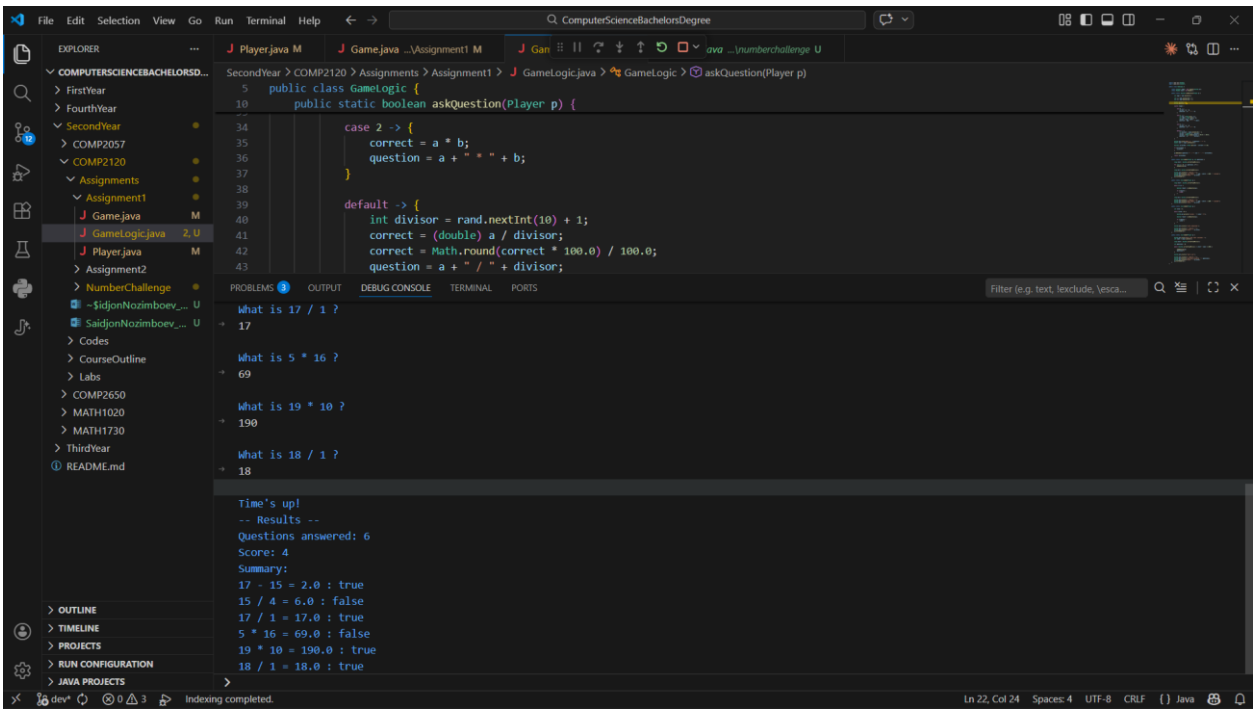
Mode3


SecondYear > COMP2120 > Assignments > Assignment1 > GameLogic.java > GameLogic > askQuestion(Player p)
public class GameLogic {
public static boolean askQuestion(Player p) {
case 2 -> {
correct = a * b;
question = a + " * " + b;
}
default -> {
int divisor = rand.nextInt(10) + 1;
correct = (double) a / divisor;
correct = Math.round(correct * 100.0) / 100.0;
question = a + " / " + divisor;
==== Game Menu ====
1) Make a wish
2) No Mistakes
3) Take Chances
4) Time Trial
Choose a mode:
3
[Lives: 3]
What is 17 / 9 ?
9
[Lives: 2]
What is 3 / 1 ?
3
[Lives: 2]
What is 17 - 13 ?
4
[Lives: 2]
What is 10 - 3 ?


SecondYear > COMP2120 > Assignments > Assignment1 > GameLogic.java > GameLogic > askQuestion(Player p)
public class GameLogic {
public static boolean askQuestion(Player p) {
case 2 -> {
correct = a * b;
question = a + " * " + b;
}
default -> {
int divisor = rand.nextInt(10) + 1;
correct = (double) a / divisor;
correct = Math.round(correct * 100.0) / 100.0;
question = a + " / " + divisor;
3
[Lives: 2]
What is 17 - 13 ?
4
[Lives: 2]
What is 10 - 3 ?
9
[Lives: 1]
What is 17 - 9 ?
3
Lives exhausted!
-- Results --
Score: 2
Summary:
17 / 9 = 9.0 : false
3 / 1 = 3.0 : true
17 - 13 = 4.0 : true
10 - 3 = 9.0 : false
17 - 9 = 3.0 : false

Mode4


SecondYear > COMP2120 > Assignments > Assignment1 > GameLogic.java > GameLogic > askQuestion(Player p)
public class GameLogic {
public static boolean askQuestion(Player p) {
case 2 -> {
correct = a * b;
question = a + " * " + b;
}
default -> {
int divisor = rand.nextInt(10) + 1;
correct = (double) a / divisor;
correct = Math.round(correct * 100.0) / 100.0;
question = a + " / " + divisor;
==== Game Menu ====
1) Make a wish
2) No Mistakes
3) Take Chances
4) Time Trial
Choose a mode:
4
Enter time limit (seconds):
10
What is 17 - 15 ?
2
What is 15 / 4 ?
6
What is 17 / 1 ?
17
What is 5 * 16 ?
69



3.

Time Limit:

The time limit uses System.currentTimeMillis(). The program records the start time and keeps asking questions until the current time passes the time limit.

Lives:

The lives system starts with 3 lives. One life is removed for each wrong answer, and the game ends when lives reach 0.

Scoring:

The score increases by 1 for each correct answer and is displayed at the end of the game.

Part B

1.

```java
public class Player {

    String name;
    int score;
    String[] summary;
    int index;

    public Player(String name, int maxQuestions) {
        this.name = name;
        this.score = 0;
        summary = new String[maxQuestions];
        index = 0;
    }

    public void addSummary(String s) {
        summary[index] = s;
        index++;
    }

    public void printSummary() {
        System.out.println("Summary:");
        for (int i = 0; i < index; i++) {
            System.out.println(summary[i]);
        }
    }
}
```

Player Class Explanation: Player class represents each player in the game. It stores the player's name, score, and a summary of answers using a string array. The constructor initializes the player's name, sets the score to 0, and creates the summary array to store the results of each question.

Competition Loop Explanation: First the program asks the user to select a game mode. Then it asks for the number of players and creates a Player object for each one. Each player enters their name and plays the game in the selected mode. After all players finish, the program sorts the players by score and displays a leaderboard.

2.

Mode1

```java
public class Player {
    String[] summary;
    int index;

    public Player(String name, int maxQuestions) {
        this.name = name;
        this.score = 0;
        summary = new String[maxQuestions];
        index = 0;
    }
```

```
1) Single Player
2) Multiplayer
Choose option:
2

==== Game Menu ====
1) Make a wish
2) No Mistakes
3) Take Chances
4) Time Trial
Choose a mode:
1

How many players?
2

How many questions?
2

Enter player 1 name:
Sam

What is 1 / 1 ?
1

What is 14 / 3 ?
```

```java
public class Player {
    String[] summary;
    int index;

    public Player(String name, int maxQuestions) {
        this.name = name;
        this.score = 0;
        summary = new String[maxQuestions];
        index = 0;
    }
```

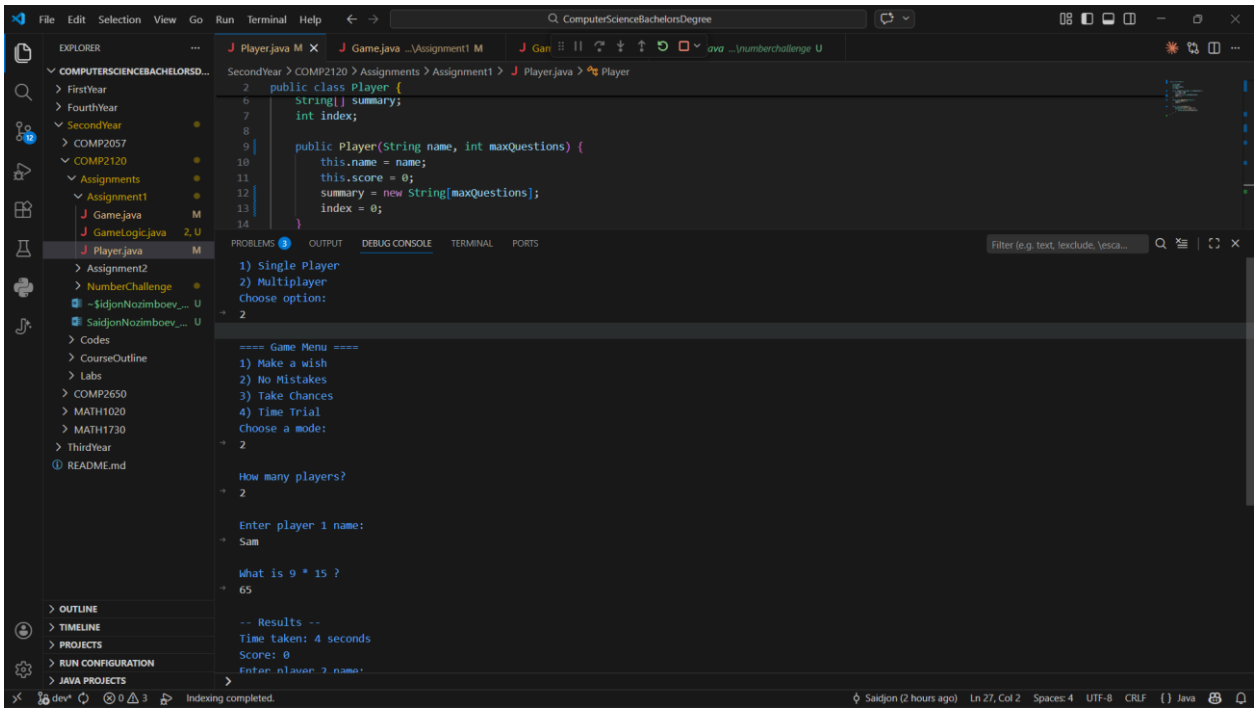
```
What is 14 / 3 ?
6

-- Results --
Time taken: 6 seconds
Score: 1
Summary:
1 / 1 = 1.0 : true
14 / 3 = 6.0 : false
Enter player 2 name:
Leo

What is 10 - 8 ?
2

What is 6 * 13 ?
78

-- Results --
Time taken: 11 seconds
Score: 2
Summary:
10 - 8 = 2.0 : true
6 * 13 = 78.0 : true
-- Leaderboard --
```

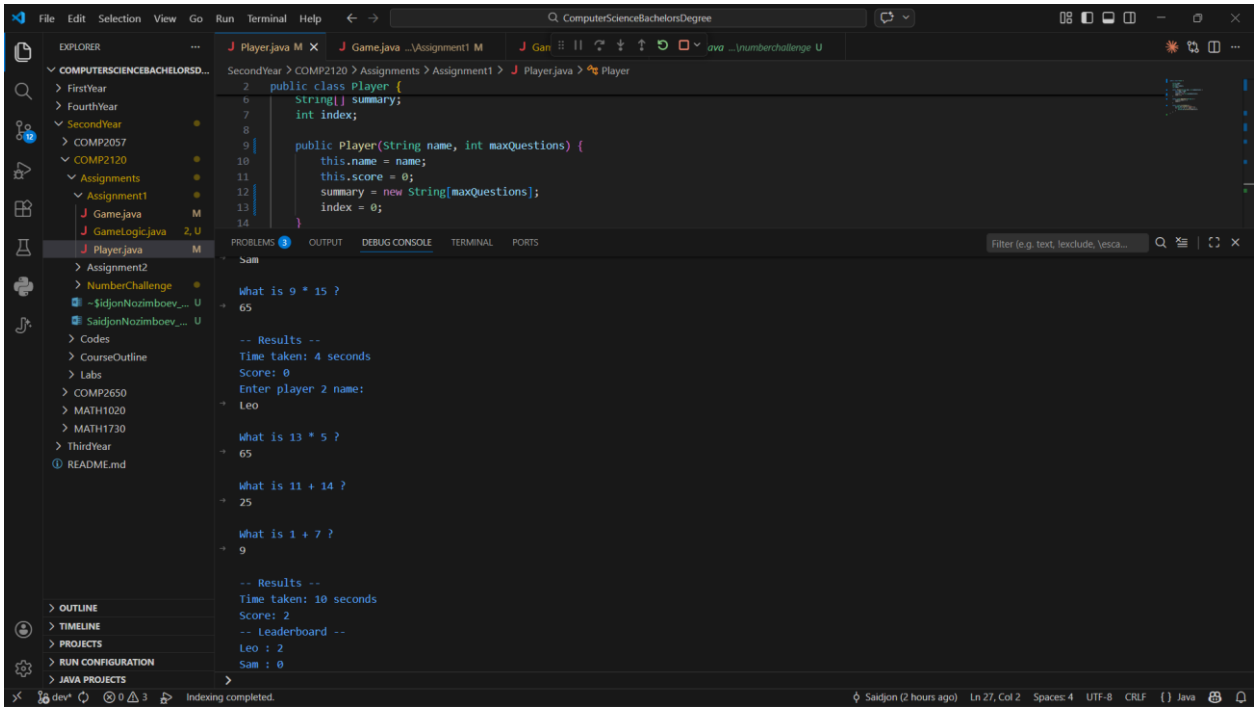

```java
public class Player {
    String[] summary;
    int index;

    public Player(String name, int maxQuestions) {
        this.name = name;
        this.score = 0;
        summary = new String[maxQuestions];
        index = 0;
    }
```

```
-- Results --
Time taken: 6 seconds
Score: 1
Summary:
1 / 1 = 1.0 : true
14 / 3 = 6.0 : false
Enter player 2 name:
Leo

What is 10 - 8 ?
2

What is 6 * 13 ?
78

-- Results --
Time taken: 11 seconds
Score: 2
Summary:
10 - 8 = 2.0 : true
6 * 13 = 78.0 : true
-- Leaderboard --
```


```
Leo : 2
Sam : 1
```

Mode2


SecondYear > COMP2120 > Assignments > Assignment1 > Player.java > Player
public class Player {
String[] summary;
int index;
public Player(String name, int maxQuestions) {
this.name = name;
this.score = 0;
summary = new String[maxQuestions];
index = 0;
1) Single Player
2) Multiplayer
Choose option:
2
==== Game Menu ====
1) Make a wish
2) No Mistakes
3) Take Chances
4) Time Trial
Choose a mode:
2
How many players?
2
Enter player 1 name:
Sam
What is 9 * 15 ?
65
-- Results --
Time taken: 4 seconds
Score: 0


SecondYear > COMP2120 > Assignments > Assignment1 > Player.java > Player
public class Player {
String[] summary;
int index;
public Player(String name, int maxQuestions) {
this.name = name;
this.score = 0;
summary = new String[maxQuestions];
index = 0;
What is 9 * 15 ?
65
-- Results --
Time taken: 4 seconds
Score: 0
Enter player 2 name:
Leo
What is 13 * 5 ?
65
What is 11 + 14 ?
25
What is 1 + 7 ?
9
-- Results --
Time taken: 10 seconds
Score: 2
-- Leaderboard --
Leo : 2
Sam : 0

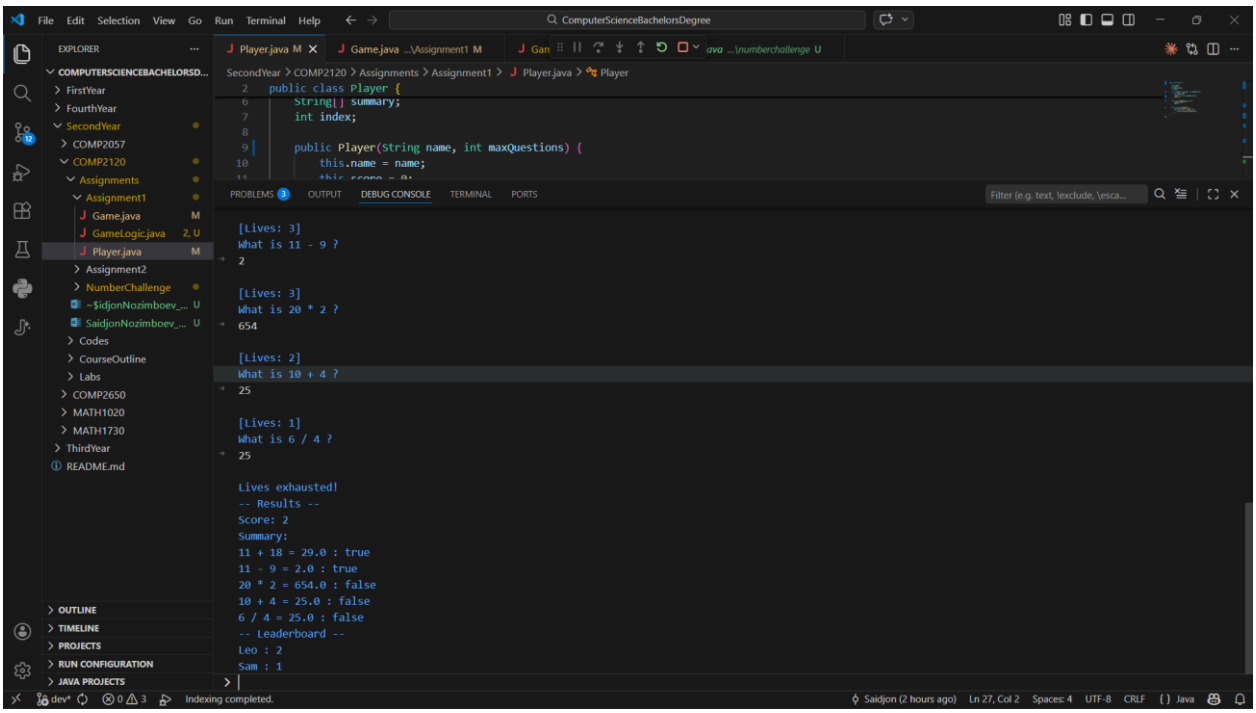
Mode3

```
public class Player {
    String[] summary;
    int index;

    public Player(String name, int maxQuestions) {
        this.name = name;
```

```
1) Single Player
2) Multiplayer
Choose option:
2

==== Game Menu ====
1) Make a wish
2) No Mistakes
3) Take Chances
4) Time Trial
Choose a mode:
3

How many players?
2

Enter player 1 name:
Sam

[Lives: 3]
What is 8 - 1 ?
7

[Lives: 3]
What is 10 / 6 ?
6

[Lives: 2]
What is 12 / 2 ?
```

```
public class Player {
    String[] summary;
    int index;

    public Player(String name, int maxQuestions) {
        this.name = name;
```

```
4

[Lives: 1]
What is 18 / 7 ?
58

Lives exhausted!
-- Results --
Score: 1
Summary:
8 - 1 = 7.0 : true
10 / 6 = 6.0 : false
12 / 2 = 4.0 : false
18 / 7 = 58.0 : false
Enter player 2 name:
Leo

[Lives: 3]
What is 11 + 18 ?
29

[Lives: 3]
What is 11 - 9 ?
2

[Lives: 3]
What is 20 * 2 ?
654
```


File Edit Selection View Go Run Terminal Help
ComputerScienceBachelorsDegree
EXPLORER
COMPUTERSCIENCEBACHELORSD...
FirstYear
FourthYear
SecondYear
COMP2057
COMP2120
Assignments
Assignment1
Game.java M
GameLogic.java 2, U
Player.java M
Assignment2
NumberChallenge
~$idjonNozimboev_... U
SaidjonNozimboev_... U
Codes
CourseOutline
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COMP2650
MATH1020
MATH1730
ThirdYear
README.md
OUTLINE
TIMELINE
PROJECTS
RUN CONFIGURATION
JAVA PROJECTS
Player.java M
Game.java ...\Assignment1 M
SecondYear > COMP2120 > Assignments > Assignment1 > Player.java > Player
public class Player {
String[] summary;
int index;
public Player(String name, int maxQuestions) {
this.name = name;
PROBLEMS 3
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
[Lives: 3]
What is 11 - 9 ?
2
[Lives: 3]
What is 20 * 2 ?
654
[Lives: 2]
What is 10 + 4 ?
25
[Lives: 1]
What is 6 / 4 ?
25
Lives exhausted!
-- Results --
Score: 2
Summary:
11 + 18 = 29.0 : true
11 - 9 = 2.0 : true
20 * 2 = 654.0 : false
10 + 4 = 25.0 : false
6 / 4 = 25.0 : false
-- Leaderboard --
Leo : 2
Sam : 1
Indexing completed.
Saidjon (2 hours ago)
Ln 27, Col 2
Spaces: 4
UTF-8
CRLF
Java

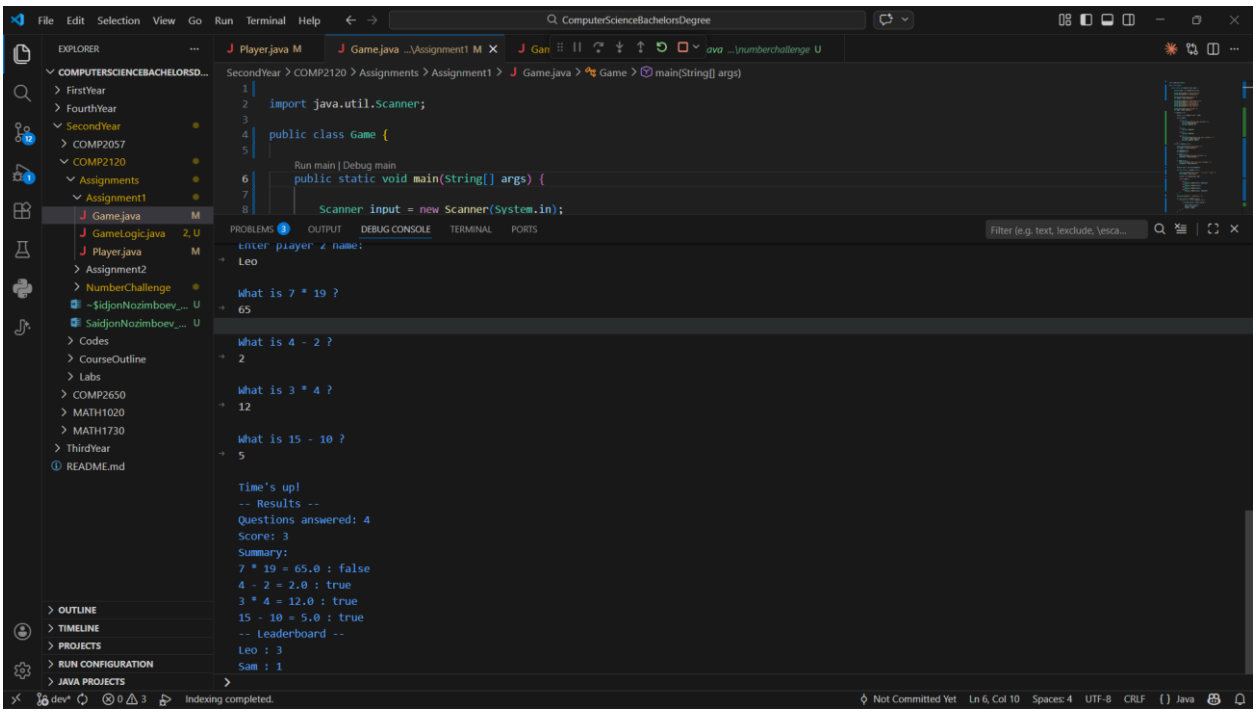
Mode4

```
import java.util.Scanner;

public class Game {

    Run main | Debug main
    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);
```

```
1) Single Player
2) Multiplayer
Choose option:
2

==== Game Menu ====
1) Make a wish
2) No Mistakes
3) Take Chances
4) Time Trial
Choose a mode:
4

How many players?
2

Enter time limit (seconds):
5

Enter player 1 name:
Sam

What is 20 - 1 ?
19

What is 15 / 8 ?
5
```

```
import java.util.Scanner;

public class Game {

    Run main | Debug main
    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);
```

```
What is 15 / 8 ?
5

What is 6 / 9 ?
6

What is 2 / 9 ?
2

Time's up!
-- Results --
Questions answered: 4
Score: 1
Summary:
20 - 1 = 19.0 : true
15 / 8 = 5.0 : false
6 / 9 = 6.0 : false
2 / 9 = 2.0 : false
Enter player 2 name:
Leo

What is 7 * 19 ?
65

What is 4 - 2 ?
2
```


File Edit Selection View Go Run Terminal Help
ComputerScienceBachelorsDegree
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COMPUTERSCIENCEBACHELORSD...
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Assignments
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Game.java M
GameLogic.java 2, U
Player.java M
Assignment2
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~$idjonNozimboev_... U
SaidjonNozimboev_... U
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PROJECTS
RUN CONFIGURATION
JAVA PROJECTS
Player.java M
Game.java ...\Assignment1 M
SecondYear > COMP2120 > Assignments > Assignment1 > Game.java > Game > main(String[] args)
import java.util.Scanner;
public class Game {
Run main | Debug main
public static void main(String[] args) {
Scanner input = new Scanner(System.in);
PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS
Leo
What is 7 * 19 ?
65
What is 4 - 2 ?
2
What is 3 * 4 ?
12
What is 15 - 10 ?
5
Time's up!
-- Results --
Questions answered: 4
Score: 3
Summary:
7 * 19 = 65.0 : false
4 - 2 = 2.0 : true
3 * 4 = 12.0 : true
15 - 10 = 5.0 : true
-- Leaderboard --
Leo : 3
Sam : 1
Indexing completed.
Not Committed Yet Ln 6, Col 10 Spaces: 4 UTF-8 CRLF Java